

## CPE 102 Computer Programming II

### Lab 7

**Part 1.** You will implement a **class named Trip**, which represents a general travel plan.

#### Instance Variables:

- destination (String): the place you are traveling to.
- days (int): number of days the trip will last.

#### Methods you must implement:

- **Constructor:** Takes destination and days as parameters and initializes the instance variables.
- **Copy Constructor**
- **String getDestination():** Accessor method.
- **int getDays():** Accessor method.
- **void setDestination(String destination):** Mutator method.
- **void setDays(int days):** Mutator method.
- **String toString():** Returns a string like:  
"Trip to destination for days days"
- **boolean equals(Trip other):** Returns true if both destination and days are equal.
  
- **Trip extend(Trip other):** This method takes another Trip object. Returns a new Trip:
  - 1) The destination becomes: "destination & other.destination"
  - 2) The days are the **sum** of both trips' durations.

Now build on Trip and create a special kind of trip: international travel. **Implement a class called InternationalTrip that extends Trip.**

#### Additional Instance Variables:

- passportRequired (boolean): whether a passport is needed.
- currency (String): the local currency at the destination.

#### Methods you must implement:

- **Constructor:** Takes 4 parameters: destination, days, passportRequired, currency.
- **String toString():** Must return something like:

International Trip to destination for days days  
 Passport Required: true/false  
 Currency: currency

You **MUST** use **super.toString()** for the first line.

- **boolean equals(IntertionalTrip other):** Returns true if destination, days, passportRequired, and currency are all equal. You MUST use super.equals().
- **InternationalTrip extend(Trip other):** Returns a new InternationalTrip:
  - 1) Destination becomes: "destination & other.destination"
  - 2) Days are added together
  - 3) Passport requirement and currency remain the same as the **calling object**

**Part 2.** Design a **class called Reservation** that represents a general reservation made by a person.

#### **Instance Variables:**

- name (String): The name of the person who made the reservation.
- dayCount (int): Number of days for the reservation.

#### **Methods you must implement:**

- **Constructor:** Takes two parameters (name, dayCount) and initializes the instance variables.
- **Copy Constructor**
- **String getName():** Accessor method.
- **int getDayCount():** Accessor method.
- **void setName(String name):** Mutator method.
- **void setDayCount(int daycount):** Mutator method.
- **String toString():** Returns a string in the format:  
 "Reservation for name: dayCount days"

- **boolean equals(Reservation other):** Returns true if both name and dayCount match.
- **Reservation extend(Reservation other):** This method takes another Reservation object and returns a new Reservation object with:
  - 1) name: Concatenation of both names as "name & other.name"
  - 2) dayCount: Sum of both day counts

Create a class called **HotelReservation** that extends the **Reservation class**, representing a hotel booking.

**Additional Instance Variables:**

- roomType (String): The type of room booked (e.g., "Single", "Double", "Suite")
- breakfastIncluded (boolean): Whether breakfast is included in the reservation

**Methods you must implement:**

- **Constructor:** Takes four parameters: name, dayCount, roomType, breakfastIncluded
- **String toString():** Returns a string in the following format:

Reservation for name: dayCount days  
 Room Type: roomType  
 Breakfast Included: true/false

You **MUST** call **super.toString()** to receive full credit.

- **boolean equals(HotelReservation other):** Returns true if all attributes match (including those in the superclass). You **must** call super.equals() correctly.
- **HotelReservation extend(Reservation other):** Returns a new HotelReservation object with:
  - 1) name: Concatenation of names
  - 2) dayCount: Sum of days
  - 3) roomType and breakfastIncluded: same as the calling object

